v158TH FIGHTER WING
HOME OF THE GREEN MOUNTAIN BOYS
THE GREEN MOUNTAIN BOYS
HQ VERMONT AIR NATIONAL GUARD

STRIKE!

Defensive Basic Fighter Maneuvers

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Basic Fighter Maneuvering (BFM)

Basic Tenets of ACM

Outlined here are the common tenets of BFM which represent the fundamentals of air combat.

- **Lose Sight, Lose the Fight**
  - You must gain and maintain a visual with the bandit throughout the engagement.

- **Maneuver in Relation to the Bandit**
  - Observe, predict, and maneuver. Fly the aircraft to take advantage of errors committed by the bandit.

- **Nose Position vs. Energy**
  - Assess your options whether to sacrifice energy advantage for WEZ advantage, and vice versa.

- **Lead vs. Pure vs. Lag Pursuit**

Defensive BFM
(DBFM)

Defensive BFM

This section will describe three primary objectives in Defensive Basic Fighter Maneuvering:

- **Defeat**
    - Outperform the adversary with potential energy and turn rate.
- **Deny**
    - Avoid acquisition or a weapons solution by the adversary.
- **Neutralize**
    - Force an effective overshoot or a successful reversal.

Defensive BFM

- Adversary Acquisition:
  - If the defender acquires the adversary outside of the turn circle, the defender should assume the adversary has visual contact, and has a positive rate of closure.
  - If the adversary pulls into any pursuit, continue with a maximum performance, high load factor turn. Turning angle prior to the adversary pulling pure or lead pursuit is more geometry for the adversary to solve.
  - As the adversary enters the defender's turn circle, they will attempt to solve closure and geometry with in and out of plane maneuvers. The defender's goal is to defeat the adversary's desired weapon solution.

Defensive BFM

- Initial Moves:
  - Once the adversary is acquired, react. The appropriate reaction depends on performance, geometry, and range. The initial move creates geometry problems for the adversary. There are three types of defensive turns:
    - Navigation: An energy sustaining turn at low load factor to assess the adversary's intentions.
    - Hard: An energy sustaining turn at moderate load factor to deny a weapons solution.
    - Break: A maximum performance turn at high load factor to defeat a weapons solution or force an overshoot.
- Defensive Cues:
  - Accurate assessment and understanding of fundamentals are the key to the success in BFM engagements. To counter an attack, Observe, Predict, Maneuver, and Assess.

Defensive BFM

- Overshoot:
  - There are two types of overshoots to take advantage of in any engagement:
    - A longitudinal overshoot, wherein the adversary crosses the tail-nose line of the defender.
    - A lateral overshoot, wherein the adversary crosses the 3/9 line of the defender. Typically seen as the traditional type of overshoot in an ACM scenario.
- Reversal:
  - Following a longitudinal overshoot, though optimally in anticipation of such, the defender should reverse their plane of motion, placing their lift vector on the adversary as they overshoot, and forcing the fight into a neutral stance, likely in the form of flat scissors.

High Aspect BFM (HABFM)

Additional Study Materials

Art of the Kill by Peter Bonanni
Introduction to High-Aspect *BFM* by Mike Solyom
Student Defensive BFM by Cider C

End

You have completed the lesson portion covering DBFM.